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Research Paper

Learners' perspective on the inclusion of gamification and advanced technology as teaching methods in undergraduate pharmacy curriculum: A one-arm post-interventional assessment

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A R T I C L E I N F O

Keywords:
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Augmented reality
Virtual reality

A B S T R A C T

Background: Gamification involves applying game design elements to non-game contexts. Pharmacy students often encounter overwhelming resources, highlighting the need for innovative strategies to enhance learning.
Objective: To assess undergraduate pharmacy students' perceptions of using gamification and advanced technology (AT) as teaching methods at the University of Nigeria Nsukka (UNN).
Methods: A one-arm post-interventional survey was conducted, using stratified random sampling to select participants from professional pharmacy classes. The intervention involved a video-based tutorial educating students on gamification and AT, followed by a Google form questionnaire to gather responses. Descriptive statistics summarized the data, while Chi-square and regression analyses identified factors influencing perceptions and outcomes.
Results: A total of 441 students participated (response rate = 72.1 %), with 271 (61.5 %) female participants and 266 (60.3 %) living off-campus. Only 26 (5.9 %) found traditional teaching methods effective, and 197 (44.7 %) reported difficulty understanding course content, with 282 (63.9 %) resorting to cramming. Prior to the intervention, 105 (23.8 %) and 222 (51.4 %) were familiar with gamification and AT concepts, respectively. After the intervention, 244 (55.3 %) agreed that gamification and 258 (58.5 %) advanced technology such as simulated learning to improve learning experiences. Gender (being male) was a predictor of a lower likelihood of achieving positive outcomes from gamification and AT in pharmacy education (OR: 0.394, 95 % CI: 0.230–0.673, $p = 0.001$).
Conclusion: Overall, pharmacy students had a fair knowledge of gamification and advanced technology, and most believed that these approaches would enhance their learning outcomes, suggesting a strong potential for integrating such methods into pharmacy education.

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Introduction

Learning is a pivotal process in human development, demanding intense concentration and discipline. Over time, educators in classrooms have developed numerous strategies to enhance students' motivation and learning performance. While stand-up assessments, rigorous training stages, and classroom competitions have been tested,[1] the waning interest suggests their inadequacy.[2] Pharmacy education mandates a continuous and rigorous learning approach. Acquiring knowledge about the specific use of medicines, the correct clinical approach to patient care, device counseling, drug formulation, preparation, and the ability to conduct quality research for drug system improvement and innovation is imperative.[3] The complexity of the pharmacy curriculum and high demands on students have resulted in diminished motivation, viewing education merely as a survival approach.[4]

A gamified learning system has been proven to be effective in enhancing learning outcomes in higher education.[5] Universities represent the pinnacle of education, focusing on nurturing highly skilled individuals, fostering intellectual growth, and shaping future leaders.[6] They serve as hubs where students gather to acquire knowledge, develop their ability to comprehend and appreciate their surroundings, and gain advanced technical expertise. Gamification is now believed to improve students' retention and extends beyond the classroom, encompassing all facets of education.

Gamification fosters character building, often overlooked in education, by promoting all-around skills. The process encourages active participation, reinforces desirability and study habits, and fosters the adaptation of specific learning behaviors and strategies.[7,8] The gamification experience in pharmacy education can be enhanced with advanced technologies (AT), creating a system where students are not just learning but are immersed or involved in the process. Advanced technology aids this process by promoting a systematic approach to learning.

The infusion of advanced technologies, such as AI, virtual reality (VR), augmented reality (AR), and machine learning (ML), signifies a transformative shift in educational approaches, opening avenues for innovative concepts like gamification.[9] Integrating artificial intelligence (AI) into pharmacy education aligns with the global commitment to quality education, a cornerstone of the United Nations Sustainable Development Goals for 2030. Emphasizing creativity as a crucial skill in education, AI can play a pivotal role in fostering this attribute among pharmacy students. Beyond conventional knowledge acquisition, AI facilitates critical thinking, problem-solving, and the generation of innovative ideas – key elements of creativity.[10] AI's contribution extends across disciplines, enabling educators to empower students to explore new perspectives, take risks, and cultivate their unique talents. Additionally, by incorporating AI to enhance creativity, pharmacy education can elevate student engagement, motivation, and overall enjoyment of the learning process.[11]

Advanced technologies make it possible to create more personalized and adaptive gamification systems. For example, AI can monitor a user's progress and adjust the game accordingly to keep it engaging, which boosts both user experience and motivation.[12] Machine learning algorithms can analyze user data to deliver tailored rewards and challenges. Combined with the immersive and interactive capabilities of VR and AR, gamification allows users to experience feedback, challenges, and rewards in real time. These technologies lay the foundation for creating dynamic, adaptive gamified systems that are more engaging, motivating, and impactful for users.[13]

Several studies have reported notably higher exam averages among game-based learners in Lebanese[14] and Australian pharmacy schools.[15] Notwithstanding the numerous benefits of gamification in pharmacy education, gamification has yet to be introduced into the Nigerian pharmacy education curriculum. More so, no study has explored the perception of pharmacy students towards the inclusion of gamification and advanced technology in the Nigerian pharmacy education curriculum.

Hence, this study aimed to assess the perception of pharmacy students at the University of Nigeria towards the inclusion of gamification and advanced technology in their curriculum.

Methods

Study design

The study was conducted as a one-arm post-interventional cohort survey among pharmacy undergraduate students at the Faculty of Pharmaceutical Sciences of the University of Nigeria, Nsukka.

Study setting

The study was conducted at the University of Nigeria, Nsukka Pharmacy School. The Faculty of Pharmaceutical Sciences started in 1967 as a Department of Pharmacy in the Faculty of Sciences. Because of the Nigerian Civil War (1967–1970), the department did not function actively until 1970. It then became part of the Faculty of Biological Sciences. The department was elevated to the status of a faculty in 1980, originally with five departments, which were later increased to six. The first set of students graduating with the declassified program passed out in the 1991/92 academic year.

Both the Bachelor of Pharmacy and Doctor of Pharmacy programs are being run concurrently. The Bachelor of Pharmacy degree is scheduled to be phased out by the end of 2024, while the Doctor of Pharmacy program started fully in 2021. It was developed based on the fundamental assumption that students ought to acquire a broad and balanced foundation in all areas of pharmaceutical sciences and practice.

Study population, sample size, and inclusion criteria

The study population comprised only students in the professional classes of pharmacy education: second, third, fourth, and fifth years of the undergraduate pharmacy program at the University of Nigeria, Nsukka. The students in the first year of study were excluded from the study because they were not fully in the pharmacy program yet and had no experience with pharmacy courses. With an estimated population of about 1500 students, a minimum sample size of 306 was needed for the study, assuming the default 5 %, 95 %, and 50 % margin of error, confidence interval, and response distribution, respectively. To boost the power of the sample size, twice the obtained size (612 students) was invited to participate in the study. However, the sample size was proportionately distributed across the participating classes.

Study instrument

A three-part video ***(See supplementary file I)*** was created for the study. Its validity and reliability were determined by experts in general education, pharmacy education, and pharmaceutical sciences, as well as among a sample of the study population that was excluded from the final study. The script of the 10-min video was face-validated for clarity, tone, and consistency to ensure it resonates with the target audience. The video was equally checked for visual effects, transitions, and audio syncing to enhance the storytelling. The video script was created from several concepts related to the features of gamification and AT, as well as their associated benefits and positive outcomes, as obtained from a literature search. The first episode of the video presented "Gamification and Augmented Reality in Pharmacy Education," using a mobile application to explain the concept. The second episode introduced the "Concept of Simulated Learning and Virtual Reality." The third episode delved into the "Concept of Blockchain" and summarized the overall concept. The second instrument used in the study was a questionnaire, whose face validity and reliability were assessed. Cronbach's alpha for Sections C and D was 0.98, indicating high internal consistency. The 46-item questionnaire ***(See supplementary file II)***, which was developed specifically to serve the purpose of this study had four sections, assessing: A) socio-demographic characteristics, B) Prior knowledge of gamification and advanced technologies and perception of the current education system, C) Perception on gamification and advanced technologies, based on the video, and D) Expected outcome on the use of gamification and advanced technology in Pharmacy education. The questionnaire was transformed into a Google Form, and the link was shared with the participants for data collection.

Study procedure

Upon obtaining informed consent from the eligible participants, they were invited to view the video centrally in the multimedia-powered seminar room of the institution's Department of Clinical Pharmacy and Pharmacy Management, each class at a time. Each part

Table 1
Sociodemographic details of the participants.

Characteristics		Frequency (N)	Percentage (%)
Age (years)	16–20	47	10.7
	21–25	328	74.4
	26–30	57	12.9
	>30	9	2.0
Year of study	2nd Year	87	19.7
	3rd Year	135	30.6
	4th Year	108	24.5
	5th Year	111	25.2
Gender	Female	271	61.5
	Male	169	38.3
Marital status	Single	420	95.2
	Married	20	4.5
Residence	Hostel	175	39.7
	Off Campus	266	60.3
Mode of entry	UTME*	321	72.8
	Direct Entry	30	6.8
	Transfers	90	20.4
Source of sponsorship	Self	12	2.7
	Parents/Guardians	375	85.0
	Scholarships	1	0.2
	Combination of 2 or more options	52	11.8
Reasons for choosing the pharmacy program	Interest/Passion	208	47.2
	Inspired by a Pharmacist I know	36	8.2
	Family/friends	49	11.1
	Only available option	23	5.2
	Others	11	2.5
	Combination of two or more of the above	114	25.6

* UTME: Unified Tertiary Matriculation Examination.

of the video was played twice for reinforcement, after which the Google Form link was shared to their WhatsApp-enabled numbers to submit their responses. As an incentive to participate, the participants were provided with free internet service during the study period. The Google form was set to receive only a single response from each respondent. The hyperlink was disabled after the scheduled period for the study.

Data management and analysis

The responses were downloaded to Microsoft Excel (2019) after the completion of data collection. The data was screened and thoroughly checked for errors to ensure that they were eligible to be analyzed, having complete socio-demographic information in addition to corresponding questions on gamification. The clean data was imported into Statistical Product and Service Solutions (SPSS) version 27 for statistical analysis. The primary findings of the study were summarized with descriptive analysis as percentages and frequencies (for categorical variables), and mean and standard error of mean (SEM) (for continuous variables). Perception and expected outcomes responses were presented in their original Likert scale, after which they were converted into continuous variables. The reference point for people who had good and poor perceptions was determined using the group median score, where good perception was defined as students' agreement to the positive benefits of the use of gamification and advanced technology, as well as their expected outcomes. Chi-square was used to compare the categorical findings from the descriptive analysis across various student classes. Logistic regression was also done to determine the predictors of some dichotomous study variables, using the first option in each independent variable as the reference.

Table 2
Prior knowledge of gamification and advanced technologies and perception of the current education system.

Statements	Responses	Frequency (Percentage)
Do you find it hard to understand the course contents in class?	No	244 (55.3)
	Yes	197 (44.7)
How often do you engage in cramming your professional pharmacy courses?	Not at all	13 (2.9)
	Rarely	57 (12.9)
	Sometimes	282 (63.9)
	All the time	89 (20.2)
Have you ever rewarded yourself for overcoming an academic challenge?	No	133 (30.2)
	Yes	308 (69.8)
How often do you participate in discussion groups?	Not at all	43 (9.8)
	Rarely	160 (36.3)
	Sometimes	220 (49.9)
	All the time	18 (4.1)
How often do you go back to revisit a course you have been taught?	Not at all	17 (3.9)
	Rarely	93 (21.1)
	Sometimes	230 (53.3)
	All the time	101 (22.9)
How would you rate the overall current mode of teaching?	Very ineffective	82 (18.6)
	Neither effective nor ineffective	130 (29.5)
	Somewhat effective	203 (46.0)
	Very effective	26 (5.9)
How often do you use software applications to learn?	Not at all	33 (7.5)
	Rarely	132 (29.9)
	Sometimes	205 (46.5)
	All the time	71 (16.1)
How familiar were you with the concept of gamification education before watching the video?	Not at all familiar	105 (23.8)
	Not very familiar	183 (41.5)
	Somewhat familiar	110 (24.9)
	Very familiar	43 (9.8)
When do you best understand courses	When pressured	51 (11.6)
	Neither pressured nor relaxed	91 (20.6)
	When relaxed	299 (67.8)
Technology concepts you were familiar with before watching the video	Blockchain	39 (8.8)
	Augmented reality	10 (2.3)
	Artificial intelligence	112 (25.4)
	Machine Learning	12 (2.7)
	Simulated Learning	36 (8.2)
	Combination of two or three other variables	222 (50.3)
	None	10 (2.3)

Results

Sociodemographic characteristics of the students

The total number of students who participated in this study was 441 participants (response rate = 72.1 %). There were 87(19.7 %), 135(30.6 %), 108(24.5 %), and 111(25.2 %) students from 200, 300, 400, and 500 levels, respectively. A total of 47 (10.7 %) students were between 16 and 20 years of age. Participants who lived in the hostel were 175(39.7 %), while 208(47.2 %) chose the pharmacy program because of interest/passion, and the majority, 375(85.0 %), were being sponsored by their parents/guardians. Other details about the socio-demographic characteristics of the participants are shown in Table 1.

Prior knowledge of gamification and advanced technologies and perception of the current education system

Students who found it hard to understand the course contents in class were 197(44.7 %), while only 89(20.2 %) engaged in cramming their professional courses *all the time*. Generally, less than one-quarter, 82(18.6 %) of the students rated the current mode of teaching as *very ineffective,* with only 43(9.8 %) being familiar with gamification education before watching the video. Half of the students, 222(50.3 %), were familiar with 2 or 3 other listed advanced technology (AT) concepts, including blockchain, augmented reality, simulated learning, artificial intelligence, and machine learning. See Table 2 for more details.

Perception of the students on the use of gamification and advanced technology

Table 3 presents the students' perception of the use of gamification and AT. *To a large extent,* students responded positively to integrating gamification and advanced technologies in learning. Specifically, 154(34.9 %) found gamification more enjoyable. In addition, 124 (28.1 %) stated that simulated learning can improve their motivation to learn. A total of 139(31.5 %) believed that virtual reality can enhance their learning experience, while 123(27.9 %) indicated that artificial intelligence and machine learning can also contribute to a better learning experience. Furthermore, 143(32.4 %) agreed that using gamification and other advanced

Table 3
Perception of respondents on the use of gamification and advanced technology.

Statements	NOA[a]	TLiE[b]	TSE[c]	TLaE[d]	TVLE[e]
	n (%)				
Gamification can help students develop problem-solving skills	9 (2.0)	38 (8.6)	144 (32.7)	119 (27)	131 (29.7)
Gamification and other advanced technologies can increase students' interest in the subject matter being taught	10 (2.3)	37 (8.4)	141 (32)	110 (24.9)	143 (32.4)
Gamification can foster a sense of competition and collaboration among students, leading to better learning outcomes	10 (2.3)	39 (8.8)	153 (34.7)	106 (24)	133 (30.2)
Gamification can help students better apply what they have learned to real-life situations	11 (2.5)	35 (7.9)	141 (32)	117 (26.5)	137 (31.1)
Gamification can help students develop leadership skills	21 (4.8)	70 (15.9)	157 (35.6)	101 (22.9)	92 (20.9)
Gamification can help students become more independent learners	10 (2.3)	39 (8.8)	153 (34.7)	113 (25.6)	126 (28.6)
Gamification can make learning more enjoyable for students	8 (1.8)	34 (7.7)	132 (29.9)	113 (25.6)	154 (34.9)
Gamification can improve students' engagement in learning	7 (1.6)	36 (8.2)	154 (34.9)	112 (25.4)	132 (29.9)
Simulated learning can increase students' motivation to learn	99 (22.4)	27 (6.1)	147 (33.3)	134 (30.4)	124 (28.1)
Artificial Intelligence and Machine Learning can enhance students' learning experience	10 (2.3)	28 (6.3)	154 (34.9)	126 (28.6)	123 (27.9)
Augmented reality can enhance students' learning experience.	9 (2.0)	43 (9.8)	155 (35.1)	116 (26.3)	118 (26.8)
Virtual reality can enhance students' learning experiences	7 (1.6)	33 (7.5)	141 (32)	121 (27.4)	139 (31.5)
Blockchain can promote active learning among students	26 (5.9)	72 (16.3)	170 (38.5)	88 (20)	85 (19.3)
Gamification and other advanced technologies can help students better retain the information learned	5 (1.1)	30 (6.8)	145 (32.9)	97 (22)	164 (37.2)
Gamification can help students better understand complex concepts	9 (2.0)	30 (6.8)	141 (32)	117 (26.5)	144 (32.7)

[a] NOA = Not At All.
[b] TLiE = To a Little Extent.
[c] TSE = To a Slight Extent.
[d] TLaE = To a Large Extent.
[e] TVLE = To a Very Large Extent.

technologies can increase their interest in the subject matter being taught. See Table 3 for more details.

Expected outcome of the use of gamification and advanced technology

Table 4 highlights students' expected outcomes on using gamification and advanced technology. Students who agreed that gamification is *extremely likely* to improve their motivation and engagement were 188(42.6 %). Generally, less than half of the students 191(43.3 %) believed that gamification in pharmacy education is *extremely likely* to improve their performance in assessments, promote critical thinking and problem-solving skills 179(40.6 %), improve their knowledge retention 199(45.1 %), and improve the overall quality of pharmacy education 187(42.4 %). See Table 4 for more details.

Factors affecting students' perception and expectations on the use of gamification and advanced technology

Table 5 presents the factors associated with students' perception and expected outcomes on the use of gamification and advanced technology. An association was observed between the frequency of cramming professional pharmacy courses and students' perception ($p = 0.012$). This implies that students who reported cramming *sometimes* 141(50.0 %) and *all the time* 58(65.2 %) had better perception of the use of gamification and advanced technology. On the other hand, an association was observed between prior familiarity with the concept of gamification and students' expected outcomes ($p = 0.048$). Students who were *somewhat familiar* 61(55.5 %) and *very familiar* 15(34.9 %) with the concept of gamification in education prior to watching the video had better expectations on the use of gamification and advanced technology. See Table 5 for more details.

Predictors of expected outcome of the use of gamification and advanced technology

The regression analysis results, presented in Table 6, showed that male students (OR: 0.359, 95 % CI: 0.209–0.617, $p \leq 0.001$) and students in their 3rd year of study (OR: 2.068, 95 % CI: 1.098–3.895, $p = 0.025$) were predictors of lower and higher likelihood of achieving expected positive outcomes from gamification and AT use in pharmacy education, respectively.

Discussion

Summary of findings

This study determined the perception of the use of gamification and advanced technology as an educational strategy among pharmacy students at the University of Nigeria. The study findings revealed that the primary driving force for students to pursue

Table 4
Expected outcome of respondents on the use of gamification and advanced technology.

Statements	Not likely	Slightly likely	Moderately likely	Very likely	Extremely likely
	N (%)				
Likelihood that gamification will improve students' motivation and engagement	16 (3.6)	35 (7.9)	90 (20.4)	112 (25.4)	188 (42.6)
Gamification in pharmacy education will improve students' performance on assessments	19 (4.3)	28 (6.3)	91 (20.6)	112 (25.4)	191 (43.3)
Gamification in pharmacy education will prepare students for future pharmacy practice	18 (4.1)	37 (8.4)	94 (21.3)	111 (25.2)	181 (41)
Gamification and advanced technologies will increase students' satisfaction with the pharmacy program	14 (3.2)	31 (7)	95 (21.5)	123 (27.9)	178 (40.4)
Gamification in pharmacy education will enhance students' engagement with course content	10 (2.3)	36 (8.2)	89 (20.2)	119 (27)	187 (42.4)
Gamification in pharmacy education will increase student interest in pursuing pharmacy education	13 (2.9)	35 (7.9)	94 (21.3)	121 (27.4)	178 (40.4)
Gamification in pharmacy education will improve student knowledge retention	16 (3.6)	30 (6.8)	77 (17.5)	119 (27)	199 (45.1)
Gamification in pharmacy education will improve the overall quality of pharmacy education	13 (2.9)	32 (7.3)	92 (20.9)	117 (26.5)	187 (42.4)
How likely do you think the use of gamification in pharmacy education will improve student perception of the pharmacy profession	16 (3.6)	38 (8.6)	85 (19.3)	119 (27)	183 (41.5)
Gamification in pharmacy education will have a positive impact on the overall pharmacy profession	16 (3.6)	39 (8.8)	80 (18.1)	124 (28.1)	182 (41.3)
Gamification in Pharmacy education will improve the learning experience	16 (3.6)	30 (6.8)	79 (17.9)	121 (27.4)	195 (44.2)
Gamification in pharmacy education will increase the level of collaboration among Pharmacy students	13 (2.9)	42 (9.5)	85 (19.3)	129 (29.3)	172 (39)
Gamification in Pharmacy education will promote critical thinking and problem-solving skills	17 (3.9)	25 (5.7)	94 (21.3)	126 (28.6)	179 (40.6)

Table 5
Factors associated with students' perception and expected outcomes on the use of gamification and advanced technology.

Statements		Perception			Expected outcome		
		Poor Perception	Good Perception	*p*-value	Poor Outcomes	Good Outcomes	*p*-value
Do you find it hard to understand course contents in class?	No	21(49.6)	123(50.4)	0.356	126(51.6)	118(48.4)	0.302
	Yes	89(45.2)	108(54.8)		92(46.7)	105(53.3)	
How often do you engage in cramming your professional pharmacy courses?	Not at all	10(76.9)	3(23.1)	0.012	9(69.2)	4(30.8)	0.356
	Rarely	28(49.1)	29(50.9)		29(50.9)	28(49.1)	
	Sometimes	141(50.0)	141(50.0)		141(50.0)	141(50.0)	
	All the time	31(34.8)	58(65.2)		39(43.8)	50(56.2)	
Have you ever rewarded yourself for overcoming an academic challenge?	No	61(45.9)	72(54.1)	0.628	71(53.4)	62(46.6)	0.276
	Yes	149(48.4)	159(51.6)		147(47.7)	161(52.3)	
How often do you participate in discussion groups?	Not at all	17(39.5)	26(60.5)	0.294	20(46.5)	23(53.5)	0.358
	Rarely	84(52.5)	76(47.5)		88(55.0)	72(45.0)	
	Sometimes	99(45.0)	121(55.0)		101(45.9)	119(54.1)	
	All the time	10(55.6)	8(44.40)		9(50.0)	9(50.0)	
How often do you go back to revisit a course you have been taught?	Not at all	10(58.8)	7(41.2)	0.754	8(47.1)	9(52.9)	0.236
	Rarely	46(49.5)	47(50.5)		54(58.1)	39(41.9)	
	Sometimes	108(47.0)	122(53.0)		112(48.7)	118(51.3)	
	All the time	46(45.5)	55(54.5)		44(43.6)	57(56.4)	
How would you rate the overall current mode of teaching?	Very ineffective	34(41.5)	48(58.5)	0.204	31(37.8)	51(62.2)	0.980
	Neither effective nor ineffective	61(46.9)	69(53.1)		64(49.2)	66(50.8)	
	Somewhat effective	98(48.3)	105(51.7)		108(53.2)	95(46.8)	
	Very effective	17(65.4)	9(34.6)		15(57.7)	11(42.3)	
How often do you use software applications to learn?	Not at all	15(45.5)	18(54.5)	0.714	16(48.5)	17(51.5)	0.921
	Rarely	68(51.5)	64(48.5)		68(51.5)	64(48.5)	
	Sometimes	96(46.8)	109(53.2)		101(49.3)	104(50.7)	
	All the time	31(43.7)	40(56.3)		33(46.5)	38(53.5)	
How familiar were you with the concept of gamification in education before watching the video?	Not at all familiar	52(49.5)	53(50.5)	0.204	58(55.2)	47(44.8)	0.048
	Not very familiar	87(47.5)	96(52.5)		83(45.4)	100(54.6)	
	Somewhat familiar	45(40.9)	65(59.1)		49(44.5)	61(55.5)	
	Very familiar	26(60.5)	17(39.5)		28(65.1)	15(34.9)	
When do you best understand courses?	When pressured	28(54.9)	23(45.1)	0.378	29(56.9)	22(43.1)	0.477
	Neither pressured nor relaxed	46(50.5)	45(49.5)		46(50.5)	45(49.5)	
	When relaxed	136(45.5)	163(54.5)		143(47.8)	156(52.2)	
Which of these technology concepts were you familiar with before watching the video	Blockchain	24(61.5)	15(38.5)	0.104	26(66.7)	13(33.3)	0.093
	Augmented reality	7(70.0)	3(30.0)		6(60.0)	4(40.0)	
	Artificial intelligence	50(44.6)	62(55.4)		59(52.7)	53(47.3)	
	Machine Learning	7(58.3)	5(41.7)		9(75.0)	3(25.0)	
	Simulated Learning	22(61.1)	14(38.9)		19(52.8)	17(47.2)	
	Combination of two or more other variables	97(43.7)	125(56.3)		94(42.3)	128(57.7)	
	None	3(30.0)	7(70.0)		5(50.0)	5(50.0)	

pharmacy as their field of study was their passion and genuine interest in the subject.

Despite the participants expressing that they did not find it challenging to grasp the course content during classes, there was a prevalent practice of cramming among them. A remarkable number of students across all years of study indicated their participation in discussion groups, demonstrating an active engagement in collaborative learning. In addition, many students reported a habit of rewarding themselves upon successfully overcoming academic challenges, suggesting the use of positive reinforcement in their learning processes. Interestingly, a substantial number of participants revealed that they revisited course materials even after the initial instruction, highlighting a commitment to reinforcing their knowledge. Moreover, participants stated their utilization of software applications as tools for learning, indicating a willingness to adopt technology-driven educational resources.

One-third of the participants exhibited familiarity with the concept of artificial intelligence, which was notably higher than their familiarity with other advanced technologies, with augmented reality ranking as the least recognized. When evaluating the participants' overall perception of these technologies based on their knowledge, usage, and application, it was intriguing to note that more respondents held a favorable perception of gamification, regardless of whether they found it difficult to understand a course or not.

However, those who indicated "not at all" regarding their engagement in cramming had a less positive perception in the study. This perception showed only a slight increase across the remaining scales, with the majority of participants found to have a very favorable perception of the subject. Respondents also had a good perception of the use of virtual reality and gamification to enhance the learning experience and foster real-life applications of knowledge. Many participants expressed strong confidence in the expected outcomes, particularly in their belief that gamification, when integrated with advanced technologies, held the potential to enhance students' interest in pursuing pharmacy education. Interestingly, those who reported feeling pressured during their learning experiences and rated the overall current mode of teaching as ineffective had a notable positive perception regarding the potential outcomes of this

Table 6
Predictors of the expected outcome on the use of gamification and advanced technology.

Characteristics		B	p-value	OR	95C·I for OR	
					Lower	Upper
Age	**16–20**	**Ref**				
	21–25	0.133	0.717	1.142	0.556	2.349
	26–30	0.052	0.919	1.054	0.386	2.877
	>30	1.163	0.203	3.201	0.534	19.196
Gender	**Female**	**Ref**				
	Male	−1.023	<0.001	0.359	0.209	0.617
Year of Study	**2nd Year**	**Ref**				
	3rd Year	0.726	0.025	2.068	1.098	3.895
	4th Year	0.566	0.129	1.762	0.848	3.662
	5th Year	0.641	0.074	1.898	0.940	3.834
Marital status	**Single**	**Ref**				
	Married	−1.370	0.010	0.254	0.090	0.720
Residence	**Hostel**	**Ref**				
	Off Campus	0.649	0.011	1.913	1.160	3.156
Mode of entry	**UTME***	**Ref**				
	Direct Entry	0.193	0.687	1.213	0.473	3.111
	Transfers	−0.500	0.078	0.606	0.347	1.058
Source of sponsorship	**Self**	**Ref**				
	Parents/Guardians	−0.062	0.925	0.940	0.259	3.413
	Scholarship	21.303	1.000	1,785,512,952.408	0.000	.
	Self, Parents/Guardians	−0.097	0.897	0.908	0.208	3.957
	Parents/Guardians; Government	0.122	0.920	1.130	0.103	12.383
	Parents/Guardians: Scholarship	0.255	0.802	1.290	0.176	9.457
	Self; Parents/Guardians; Scholarship	0.488	0.613	1.630	0.245	10.827
	Self; Parents/Guardians; Scholarship; Government	21.368	1.000	1,906,093,029.177	0.000	-.
Reason for choosing pharmacy	**Interest/Passion**	**Ref**				
	Inspired by a Pharmacist I know	−0.571	0.144	0.565	0.262	1.216
	Family/Friends	−0.395	0.257	0.674	0.341	1.334
	Only available option	−0.606	0.230	0.546	0.203	1.469
	Others	−0.202	0.760	0.817	0.223	2.993
	Two or more reasons for choosing Pharmacy	21.487	0.999	3.906	1.043	14.627
How familiar were you with the concept of gamification in education before watching the video?	**Not at all familiar**	**Ref**				
	Not very familiar	0.392	0.118	1.481	0.906	2.421
	Somewhat familiar	0.367	0.195	1.444	0.828	2.517
	Very familiar	−0.342	0.376	0.710	0.333	1.516
Which technology concepts were you familiar with before watching the video	**None**	**Ref**				
	Blockchain	−0.643	0.375	0.526	0.127	2.179
	Augmented reality	−0.526	0.563	0.591	0.099	3.520
	Artificial intelligence	−0.142	0.831	0.868	0.236	3.196
	Machine Learning	−1.059	0.257	0.347	0.055	2.168
	Simulated Learning	−0.141	0.846	0.868	0.209	3.605
	Combination of two or more other variables	0.268	0.682	1.308	0.363	4.715

*UTME: Unified Tertiary Matriculation Examination.

approach. They believed it could greatly enhance their satisfaction with pharmacy education and improve their performance on assessments. Furthermore, participants held a high regard for the potential impact of gamification on their level of collaboration, motivation, and engagement in the learning process. A substantial number of participants also expressed the belief that gamification was extremely likely to prepare students for the future, equipping them with the necessary skills to apply what they had learned in real-life scenarios. These additional findings underscore the participants' optimism regarding the positive effects of gamification and advanced technologies on their educational experiences and prospects in the field of pharmacy.

Comparison with other studies and policy implications

In the uncharted landscape of academic research, the intersection of gamification and advanced technologies remains a focus. A shift in students' priorities and timelines for deadlines across various study years has been reported in some studies,[16] with many revealing that students resort to cramming, especially when facing extensive material and time constraints in professional pharmacy courses. Cramming has been associated with short-term gains, akin to a gambling strategy, emphasizing its prevalence.[17] Siagian's perspective on cramming emphasizes the lack of long-term retention and cognitive performance decline associated with last-minute reading-induced stress.[18] Students reward themselves for overcoming challenges, reflecting Deci's concept of intrinsic motivation, which correlates with academic success.[19] The importance of continual knowledge gain is highlighted, with the Wegmans School of Pharmacy study revealing a decline in scores after a year, possibly due to long-term memory effects.[20] This emphasizes effective instruction and laboratory practices for retention.

However, poor comprehension, indicated by the study conducted among pharmacy students at Obafemi Awolowo University, leads over 50 % of students to withdraw from pharmacy school due to academic pressure. The paradigm shift to online learning during the COVID-19 pandemic, as noted by Elnour et al., reflects changes in students' perceptions and usage of technology.[21] Mills et al.'s research noted varied results based on motivation, satisfaction, confidence, and preparedness,[22] while Kahveci's study contradicts uniform comfort with technology usage across academic levels.[16]

Various aspects of student learning behaviors, preferences, and the impact of technology on education were identified in this study. Stress's negative impact on comprehension test scores is highlighted by the study of Fatemi et al.,[23] as they associated stress with the attentional control theory. At the same time, Eysenck et al. challenged this theory, noting improved performance under stress for some individuals.[24] The findings of Vogel & Schwabe support this, suggesting that memories can be enhanced if connected to the stressor.[25]

Additionally, the study underscores motivation's role in student engagement and learning experiences, aligning with Deci's concept of intrinsic motivation.[18] The impact of technology on learning habits is explored, with students' familiarity and usage dependent on exposure, socio-economic factors, knowledge, and perceived usefulness of software applications, as noted by Ogunbodede.[26] Moreover, the research delves into the influence of lecturers and tutors on students' technology usage. Students tend to observe their educators and may align their preferences based on the resources used by instructors. This aligns with the findings of Makura, suggesting that students are more accustomed to generic items like photocopying machines and projectors.[11]

The study discussed notable advanced technologies like AI, blockchain, ML, AR, and VR. AI and blockchain gain familiarity due to technologies like ChatGPT, crypto, and forex trading, while ML faces awareness challenges due to its technical nature. Simulated learning, focusing on pharmacy students at the University of Nigeria, Nsukka, aligns with Joseph et al.'s findings, indicating its potential to enhance clinical practice experiences.[27] Exploring augmented reality, the study notes that familiarity depends on exposure and knowledge. This aligns with Irhebhude et al.'s positive students' responses to augmented reality.[28]

Numerous studies have highlighted gender differences in digital game-based learning. Our research similarly found that male students are less likely to achieve the anticipated positive outcomes from the use of gamification and AT in pharmacy education. These results align with findings by Kahn et al.,[29] which indicate that boys generally prefer action-oriented games, while girls tend to favor simulation, virtual, and puzzle-based games. This distinction is believed to stem from differing motivational factors—boys often seek to dominate gameplay, focusing on mastery and competition.[28] In contrast, studies by Klisch et al.[30] and Chang et al.[31] show that female students tend to outperform their male counterparts in terms of engagement, learning, and academic achievement through game-based methods. Additionally, women often perceive gamified experiences as more enjoyable and playful, driven by their greater interest in features like badges and an emphasis on user-friendliness and enjoyment.[32]

These gender-based differences pose a challenge for gamification designers, highlighting the need for strategies that support greater gender equity in educational game design. Furthermore, women's decisions to adopt information technologies are frequently influenced by social factors, such as perceived social pressure, particularly during the early stages of adoption.[33] In the context of modern online platforms with integrated social features (e.g., social media), women are generally more socially driven in their usage, whereas men tend to focus more on functional or practical aspects.[34,35] However, in certain settings like mobile learning, it has been observed that social factors tend to influence men more than women.[36]

Additionally, the study findings showed that students in their third year of study had a higher likelihood of achieving expected positive outcomes from gamification and AT use in pharmacy education. This appreciation of the importance of gamification and AT use among the third-year pharmacy students could be due to the introduction of core pharmacy courses like pharmacology in this year of study. With the incorporation of gamification and the use of advanced technologies in teaching pharmacology, students in this year of study and beyond can better understand pharmacological terminologies such as drugs, indications, pathophysiology, and the mechanism of action of drugs. For instance, escape room activity, which is a gamification concept and commonly used in Pharmacy education, has been proven to be an effective way of learning new information related to diseases because of its positive effects on students' cognitive skills and critical thinking.[37]

In summary, the research thoroughly explores factors shaping students' learning experiences, emphasizing the role of experiential learning, gamification, and simulated learning in enhancing pharmacy education and practice. The positive impact of collaboration and gamification on student engagement, communication skills, and motivation is affirmed by various studies, underlining the importance of thoughtful implementation of gamification policies. Based on the study findings, a quick adoption and inclusion of gamification concepts and other advanced technologies into Nigerian pharmacy education must be done. With the associated benefits of gamification and other AT concepts, the numerous limitations of the current traditional teaching technique can be addressed, thus leading to the production of pharmacists attuned with the modern technology know-how.

Our study has several limitations. Firstly, due to the cross-sectional nature of this study, it was not feasible to provide students with an extended period to fully experience and evaluate the gamification application before its usage. Additionally, cross-sectional studies are generally influenced by subjective opinions; thus, some of the students' responses might be overstated. Nevertheless, the strength of this study lies in the fact that it is the first study on the perspective of Nigerian pharmacy students on the use of gamification and other advanced technologies as a teaching method in undergraduate pharmacy curriculum and thus justifies future longitudinal studies that can explore deeper into the long-term impact of gamified learning on pharmacy students' academic performance and motivation. The study had no control arm, but it was expected that their prior experience with the traditional methods in use in the school would serve as a control for all the participants. Also, our study suffered from researcher bias due to the design of the survey questions (the wording of the questions), which might have led the participants to respond in a particular way. However, there are not many disadvantages to the use of gamification and advanced technology in pharmacy education.

Conclusion

Pharmacy students had a positive perception of using gamification in their education, seeing it as a tool to boost academic performance. Frequent cramming influenced their views on gamification and other advanced technologies (AT). Male students showed strong optimism about its impact on outcomes. While it is acknowledged that integrating advanced technologies may pose challenges, both in terms of cost and complexity, the potential benefits far outweigh the expected limitations.

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Contribution to literature/Highlight

- This manuscript is the first to assess Nigerian pharmacy students' perceptions of gamification and advanced technology in education.
- It addresses a gap in the literature, as these innovative methods have not been explored in Nigeria's pharmacy curriculum.
- The study shows that using technologies like augmented reality and artificial intelligence can boost student engagement, motivation, and learning outcomes.
- Findings suggest that integrating these methods could improve the quality of pharmacy education, especially in low-resource settings.
- Adopting these strategies may better equip future pharmacists to meet the demands of modern healthcare systems.

Declaration of competing interest

The authors have no conflicting interest to declare.

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Appendix A. Supplementary data

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